Alessandro **N. Garritano | PhD**

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Biography

Alessandro Garritano is a microbial ecologist at the University of Sydney, where he studies how soil microbes store carbon and phosphorus and how that storage shapes nutrient cycling under climate extremes. He combines stable isotope labelling with amplicon and metagenome sequencing, and he writes the pipelines his group needs, including HiFiTaxa for PacBio amplicon data and AmplinkOne for downstream analysis in R.

His PhD at UNSW Sydney (2025) asked how deep sea sponges feed in the dark. Working on sponges collected off Rio de Janeiro, he showed that ammonia oxidising archaea dominate their microbiomes, fix carbon with energy from the host's ammonia and appear to be inherited within each sponge species, while bacterial partners break down the tough carbon of marine snow. To find that autotrophy in the first place he built hidden Markov models for the key carbon fixation enzymes and tested them across 52,515 bacterial and archaeal genomes, which turned up the capacity to fix carbon in lineages thought to be strictly heterotrophic. He holds an adjunct fellowship at UNSW, where he continues to collaborate on marine symbioses.

Before moving to Australia he completed a BSc in biological sciences at the University of Campinas and an MSc in microbiology at the Federal University of Rio de Janeiro, where his thesis on hydrogen and methane from palm oil mill effluent won the Chemistry Institute's best master thesis award. He is based in the School of Life and Environmental Sciences and jointly supervises two PhD students and one Honours student.

Research Experience

Aug 2025–Present
Postdoctoral Research Associate
The University of Sydney · Sydney, Australia
Microbial carbon, nitrogen and phosphorus cycling in agricultural soils. Stable isotope incubations with amplicon and metagenome sequencing of the responding communities, daily supervision of two PhD students and one Honours student, and a School funded two day amplicon sequencing workshop.

Dec 2025–Present
Adjunct Fellow
University of New South Wales · Sydney, Australia
Appointment arising from active collaborations with three UNSW research groups across the Schools of Biotechnology and Biomolecular Sciences and of Biological, Earth and Environmental Sciences.

Education

2020–2025 **PhD**, University of New South Wales, Sydney. Thesis: *Microbial biogeochemical cycles within deep-sea sponges* (UNSW Scientia PhD Scholarship)

2015–2017 **MSc in Microbiology**, Federal University of Rio de Janeiro, Brazil. Thesis: *Biological production of hydrogen and methane via anaerobic fermentation using POME hydrolysate obtained with an enzyme preparation from Ricinus communis* (Best Master Thesis of the Chemistry Institute, 2018)

2010–2014 **BSc in Biological Sciences**, State University of Campinas, Brazil. Major in molecular biology

Publications

Metrics: h-index 7 | 355 citations (Google Scholar, October 2026) | 13 peer reviewed papers, seven as first or senior author | Field Weighted Citation Impact 1.65 | three papers in the top 5% of all outputs tracked by Altmetric, with coverage by BBC News, Phys.org and G1 (Brazil) | one paper cited in a WIPO patent application (WO-2025074280-A1)

1. **A. N. Garritano**, L. J. Hill, B. Ribeiro, T. Damasceno, *et al.*, "Ammonia oxidation and recalcitrant carbon degradation fuel mixotrophic growth in the symbiont community of a deep-sea sponge," *Microbiome*, vol. 14, no. 1, p. 153, 2026. 10.1186/s40168-026-02411-y.
2. T. Thomas and **A. N. Garritano**, "Symbiotic Ammonia Oxidation in the Marine Environment," *Annual Review of Marine Science*, vol. 18, no. 1, pp. 409–427, 2026. 10.1146/annurev-marine-040824-032008.
3. M. A. Allen, **A. N. Garritano**, R. S. Peixoto, K. Busch, *et al.*, "The novel, wide-spread sponge-associated phylum *Spongivicinota* shows metabolic complementarity to its sister-phylum *Latescibacterota*," *ISME Communications*, vol. 6, ycag317, 2026. 10.1093/ismeco/ycag317.
4. F. Modolon, **A. N. Garritano**, P. Hugenholtz, C. R. Voolstra, *et al.*, "Resolving the evolutionary duality of marine symbionts: Re-defining the genus *Endozoicomonas* and proposing *Neoendozoicomonas* gen. nov.," *ISME Communications*, vol. 6, no. 1, ycag123, 2026. 10.1093/ismeco/ycag123.
5. L. J. Hill, T. Damasceno, **A. N. Garritano**, F. Modolon, *et al.*, "Contrasting microbiomes in deep-sea scleractinian and octocoral species from the South Atlantic Ocean," *Sustainable Microbiology*, vol. 3, no. 4, qvag048, 2026. 10.1093/sumbio/qvag048.
6. F. Modolon, **A. N. Garritano**, L. J. Hill, G. Duarte, *et al.*, "Putative promiscuous symbionts in deep-sea corals and crinoids may contribute to nitrogen cycling," *Microbiome*, vol. 13, no. 1, p. 234, 2025. 10.1186/s40168-025-02229-0.

7. **A. N. Garritano**, Z. Zhang, Y. Jia, M. A. Allen, *et al.*, "Simple Porifera holobiont reveals complex interactions between the host, an archaeon, a bacterium, and a phage," *The ISME Journal*, vol. 18, no. 1, wrae197, 2024. 10.1093/ismejo/wrae197.
8. B. Ribeiro, **A. N. Garritano**, I. Raimundo, N. Delgadillo-Ordoñez, *et al.*, "Not only for corals: Exploring the uptake of beneficial microorganisms for corals by sponges," *npj Biofilms and Microbiomes*, vol. 10, no. 1, p. 125, 2024. 10.1038/s41522-024-00584-8.
9. L. J. Hill, C. S. M. D. A. Messias, C. L. S. Vilela, **A. N. Garritano**, *et al.*, "Bacteria associated with the in hospite Symbiodiniaceae's phycosphere," *iScience*, vol. 27, no. 4, p. 109 531, 2024. 10.1016/j.isci.2024.109531.
10. **A. N. Garritano**, M. E. Majzoub, B. Ribeiro, T. Damasceno, *et al.*, "Species-specific relationships between deep sea sponges and their symbiotic *Nitrosopumilaceae*," *The ISME Journal*, vol. 17, no. 9, pp. 1517–1519, 2023. 10.1038/s41396-023-01439-4.
11. **A. N. Garritano**, W. Song, and T. Thomas, "Carbon fixation pathways across the bacterial and archaeal tree of life," *PNAS Nexus*, vol. 1, no. 5, pgac226, 2022. 10.1093/pnasnexus/pgac226.
12. **A. N. Garritano**, M. D. O. Faber, L. R. V. De Sá, and V. S. Ferreira-Leitão, "Palm oil mill effluent (POME) as raw material for biohydrogen and methane production via dark fermentation," *Renewable and Sustainable Energy Reviews*, vol. 92, pp. 676–684, 2018. 10.1016/j.rser.2018.04.031.
13. **A. N. Garritano**, L. R. V. De Sá, É. C. G. Aguieiras, D. M. G. Freire, *et al.*, "Efficient biohydrogen production via dark fermentation from hydrolized palm oil mill effluent by non-commercial enzyme preparation," *International Journal of Hydrogen Energy*, vol. 42, no. 49, pp. 29 166–29 174, 2017. 10.1016/j.ijhydene.2017.10.025.

Accepted and Under Review

1. M. A. Allen, **A. N. Garritano**, V. L. Fraser, R. S. Peixoto, K. Busch, B. Slaby, U. Hentschel and T. Thomas, "Metagenome-assembled genomes and viral sequences from deep-sea sponges," *Scientific Data*, accepted, 2026.
2. **A. N. Garritano**, F. Modolon and C. R. Warren, "HiFiTaxa: a reproducible analysis pipeline for full length 16S rRNA gene and ITS amplicon sequencing from PacBio reads," bioRxiv preprint, November 2026, doi: 10.64898/2026.11.03.763332. Under review at *ISME Communications*.
3. **A. N. Garritano**, C. Warren, A. Holmes, S. Liu and J. Li, "AmplinkOne: a reproducible and modular R workflow for downstream analysis of microbial marker-gene amplicon sequencing data," under review at *ISME Communications*, 2026.
4. T. Laird, K. K. Y. Tan, D. Z. Tribbia, H. S. Aronson, A. Nawrocki, **A. N. Garritano**, *et al.* (senior author B. C. Ferrari), "An inducible atmospheric chemosynthesis pathway supports growth during prolonged carbon starvation," under review, 2026.
5. L. Roager, M. D. Schostag, **A. N. Garritano**, L. Gram and S. Egan, "Dual transcriptomics reveal commensal interactions between microalgae and phycosphere bacteria," bioRxiv preprint, February 2026, doi: 10.64898/2026.02.09.704904. Under review.

Awards

2024	JAMS x ASM Student Special 2024 Judge's Award for Best Oral Presentation, Sydney, Australia
2023	Best Oral Presentation, Animal–Microbe Symbiosis Gordon Research Seminar, Lucca, Italy
2023	Outstanding Centre for Marine Science & Innovation Oral Presentation, UNSW Sydney
2021	Outstanding Centre for Marine Science & Innovation Oral Presentation, UNSW Sydney
2018	Medalha Octávio Augusto Ceva Antunes, Best Master Thesis of the Chemistry Institute, Federal University of Rio de Janeiro, Brazil

Competitive Grants, Funding & Scholarships (total ∼ AUD 566,000 equivalent)

2024	Gordon and Betty Moore Foundation Research Training Grant (USD 15,000)
2023	Animal–Microbe Symbiosis Gordon Research Seminar travel award (USD 270)
2022	International Society for Microbial Ecology travel grant (EUR 800)
2021	UNSW Scientia PhD Scholarship (AUD 176,000), Scientia Career Development Plan (AUD 40,000) and Tuition Fee Scholarship (AUD 240,000)
2015	Brazilian Ministry of Science and Technology Master Scholarship (BRL 36,000)

Invited and Conference Presentations

2026	Invited talk, Annual General Meeting of the Australian Society for Microbiology NSW/ACT Branch, Sydney, Australia
2025	Oral presentation and session co-chair, Microbial Solutions for a Changing World, Monash University, Australia
2024	Oral presentations at the Australian Microbial Ecology Conference (Sydney), the International Society for Microbial Ecology Conference (ISME19, Cape Town) and the JAMS x ASM Student Special (Sydney)
2023	Invited oral presentation, Animal–Microbe Symbiosis Gordon Research Conference, and oral presentation at the associated Gordon Research Seminar, Lucca, Italy
2022	Speed talk, International Society for Microbial Ecology Conference (ISME18), Lausanne, Switzerland

Supervision

Higher Degree Research and Honours supervision covering day to day bioinformatics training, experimental design, thesis writing and early career development.

2026–Present	**Rachel Yamamoto**, PhD (jointly supervised). Nutrient storage and competition in microbial communities in an arid pulse driven ecosystem
2025–Present	**Hao Cui**, PhD (jointly supervised). Microorganisms regulate biomass stoichiometric C:P through community composition and individual plasticity. First author poster at ISME20
2026–Present	**Ata Sohrabi**, Honours (jointly supervised). Impact of chickpea and rhizobia symbiosis on rhizosphere bacterial communities and nutrient cycling
2020–2025	A Master's and an Honours student at UNSW (jointly supervised), and bioinformatics support for 11 PhD students and five postdoctoral researchers across five research groups
2015–2017	Two undergraduate research students at the Federal University of Rio de Janeiro (jointly supervised), both graduating with First Class Honours

Teaching

2026	**Workshop Coordinator, School of Life and Environmental Sciences, The University of Sydney.** Designed the full curriculum of a two day hands on *16S & ITS Amplicon Sequencing Workshop* for 20 Honours and PhD students, developed all teaching materials and computational exercises, won School funding and delivered the program with three academic co-instructors
2022–2025	**Tutor and Laboratory Demonstrator, UNSW Sydney.** Taught more than 250 undergraduate students across molecular cell biology, advanced biochemistry and applied molecular biology, marked written and oral assessments and provided structured feedback on student reports, receiving unsolicited written commendations from students

Academic Service & Leadership

- **Elected Committee Member, Australian Society for Microbiology, NSW/ACT Branch.** Produces the quarterly branch newsletter reaching about 350 members and led a complete redesign of its format. ASM Communication Ambassador
- **Organising Committees, Joint Academic Microbiology Seminars (JAMS).** Local and national committees (Sydney 2025, Brisbane 2026). Runs the society's social media, platforming early career researchers
- **Session Co-Chair.** Microbial Solutions for a Changing World symposium, Monash University, 2025
- **Peer Review.** The ISME Journal, Microbiome, Environmental Microbiome

Outreach, Engagement & Inclusion

- **Communications Director, ABRAÇA (Associação Brasileira de Cientistas na Austrália).** Developing scholarship funded exchange pathways between Brazilian and Australian universities and hosting two Brazilian students at the University of Sydney
- **Equity, diversity and inclusion grounded in lived experience.** First generation academic from Brazil working in a second language across two countries and three research systems, mentoring international and ESL students through the transitions he navigated himself
- **International fieldwork.** Sole group representative on deep sea research expeditions, initiating collaborations that produced five coauthored publications
- **Science writing.** First author article in The Conversation, "Deep-sea sponges survive in complete darkness in ways we didn't know before" (May 2026), published in English and Portuguese
- **Media.** Research featured by BBC News, Phys.org and G1 (Brazil's largest news portal)